Universitatea POLITEHNICA din București
Facultatea de Automatică și Calculatoare

Electronică digitală

Curs – Modulul 2

Circuite pasive în regim de comutație

dr. ing. Răzvan Tătăroiu
razvan.tataroiu@upb.ro

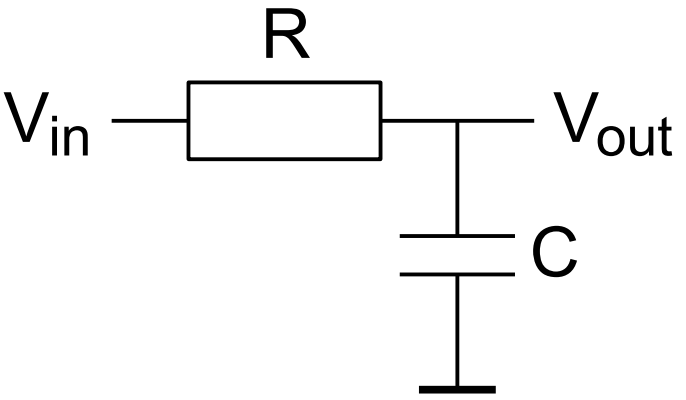
2022-2023

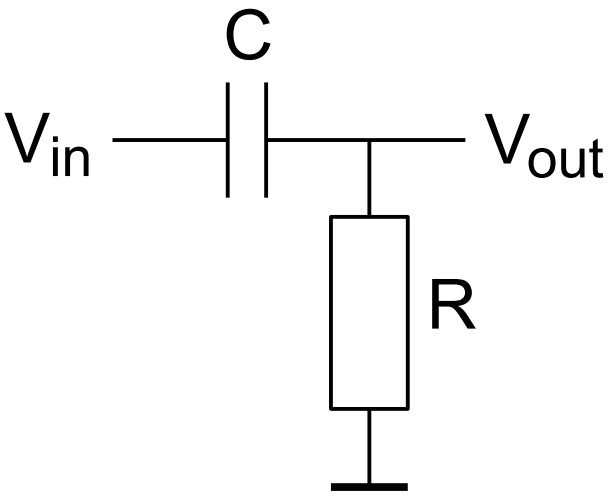
Sumar

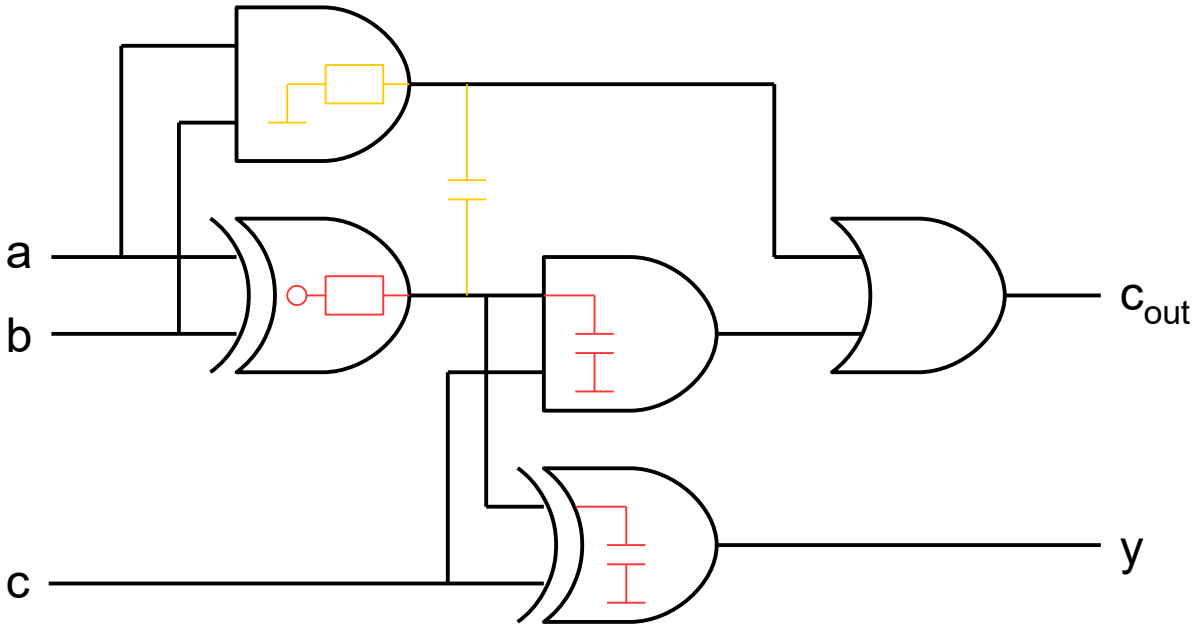
- Circuite R-C
  - intenționate
  - parazite
- Circuite cu diode

Circuite R-C – exemple

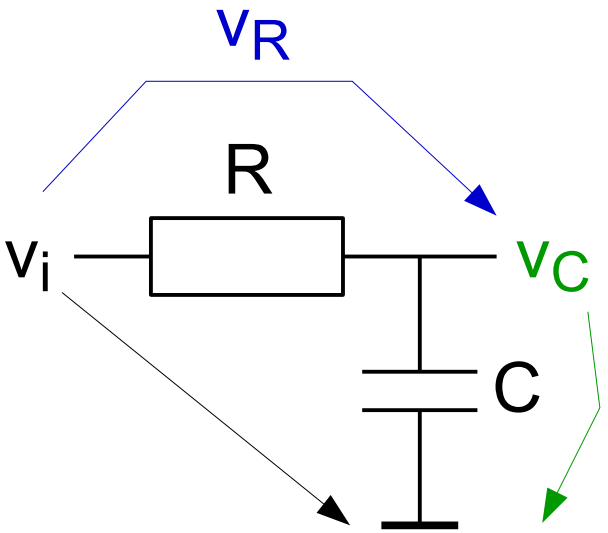
- intenționate
- parazite

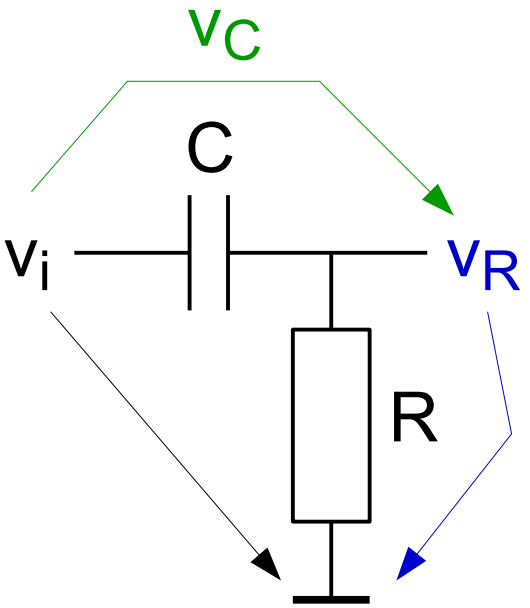






Circuite R-C – analiză (1)

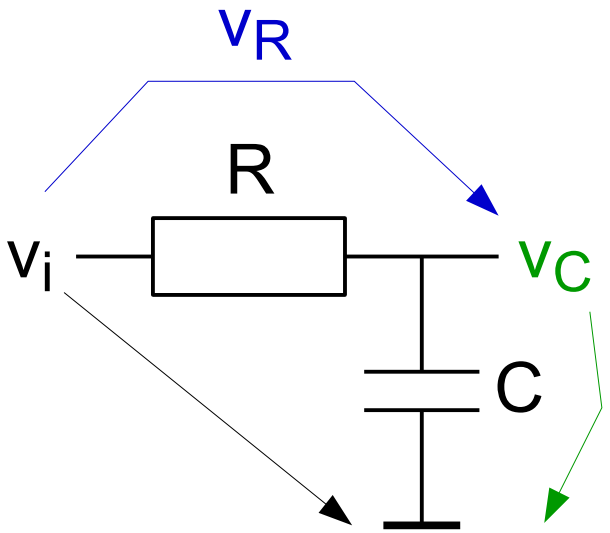




pentru ambele exemple:

$$v_i = v_R + v_C$$

Circuite R-C – analiză (2)



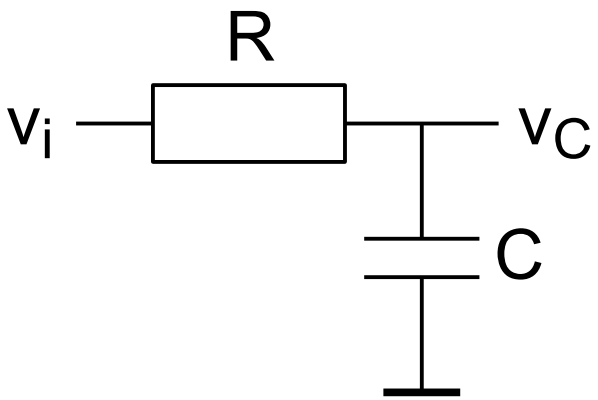
$$v_i = v_R + v_C$$

$$i = \frac{v_R}{R} = \frac{v_i - v_C}{R}$$

$$\frac{dv_C}{dt} = \frac{i}{C}$$

$$\frac{dv_C}{dt} = \frac{v_i - v_C}{RC}$$

Circuite R-C – analiză (3)



$$\frac{dv_C}{dt} = \frac{v_i - v_C}{RC}$$

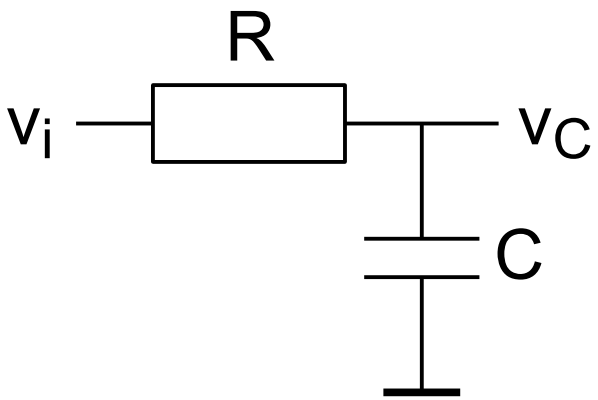
$$\tau = RC$$

constantă de timp
a circuitului RC

$$\frac{dv_C}{dt} = \frac{v_i - v_C}{\tau}$$

OBS.:
Când condensatorul s-a terminat de încărcat / descărcat
(tensiunea pe C nu mai variază) curentul este zero
=> $v_i = v_C$
* regimurile tranzitorii pe ieșire se termină în v_i

Circuite R-C – analiză (4)



unde tinde ieșirea

unde se află ieșirea

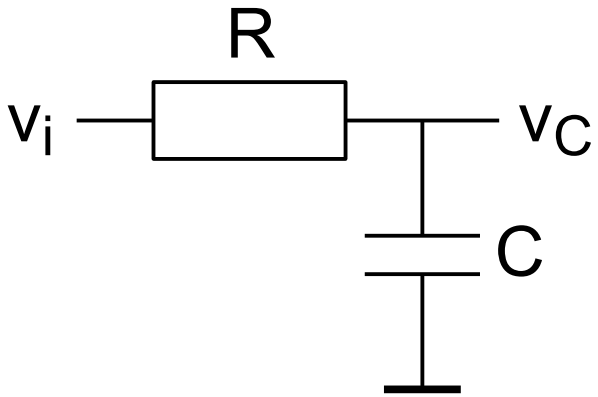
$$\frac{dv_C}{dt} = \frac{v_i - v_C}{\tau}$$

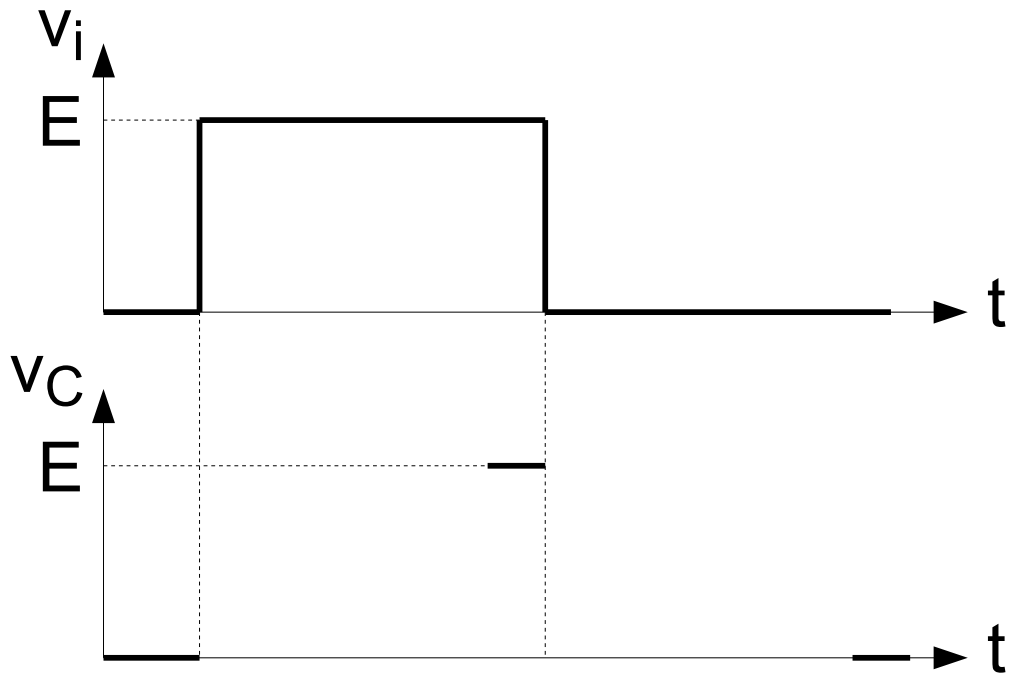
cât mai are ieșirea de parcurs

viteza de creștere a ieșirii

constanta de timp

Circuite R-C – analiză (5)

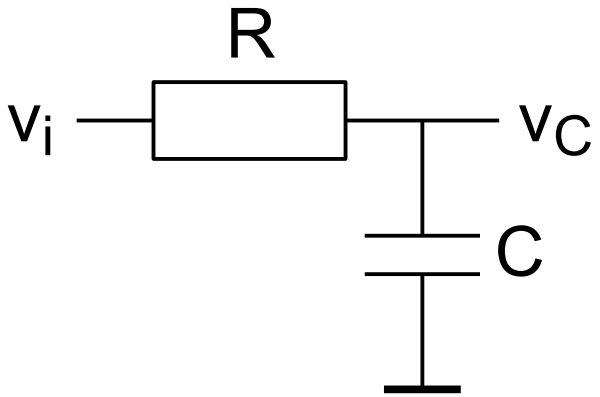


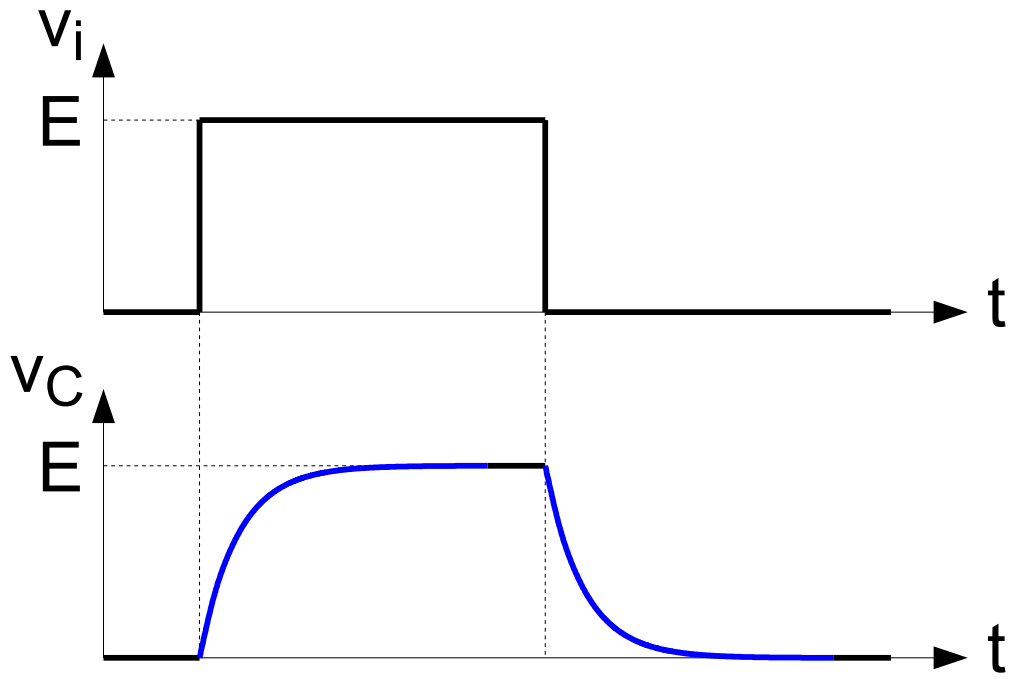


$$\frac{dv_C}{dt} = \frac{v_i - v_C}{\tau}$$

regimurile tranzitorii pe ieșire se termină în v_i

Circuite R-C – analiză (6)

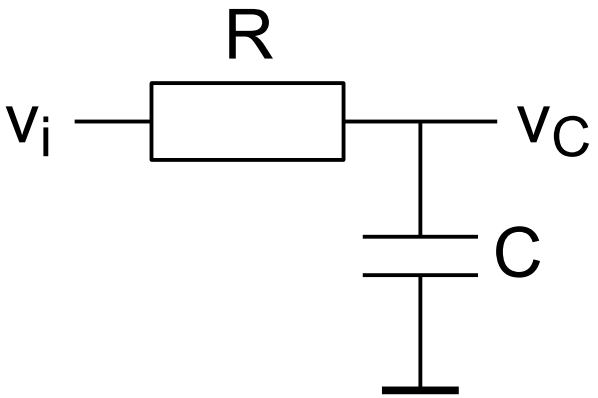


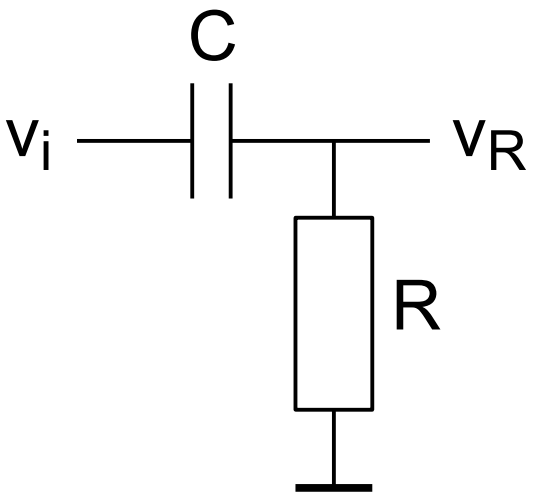


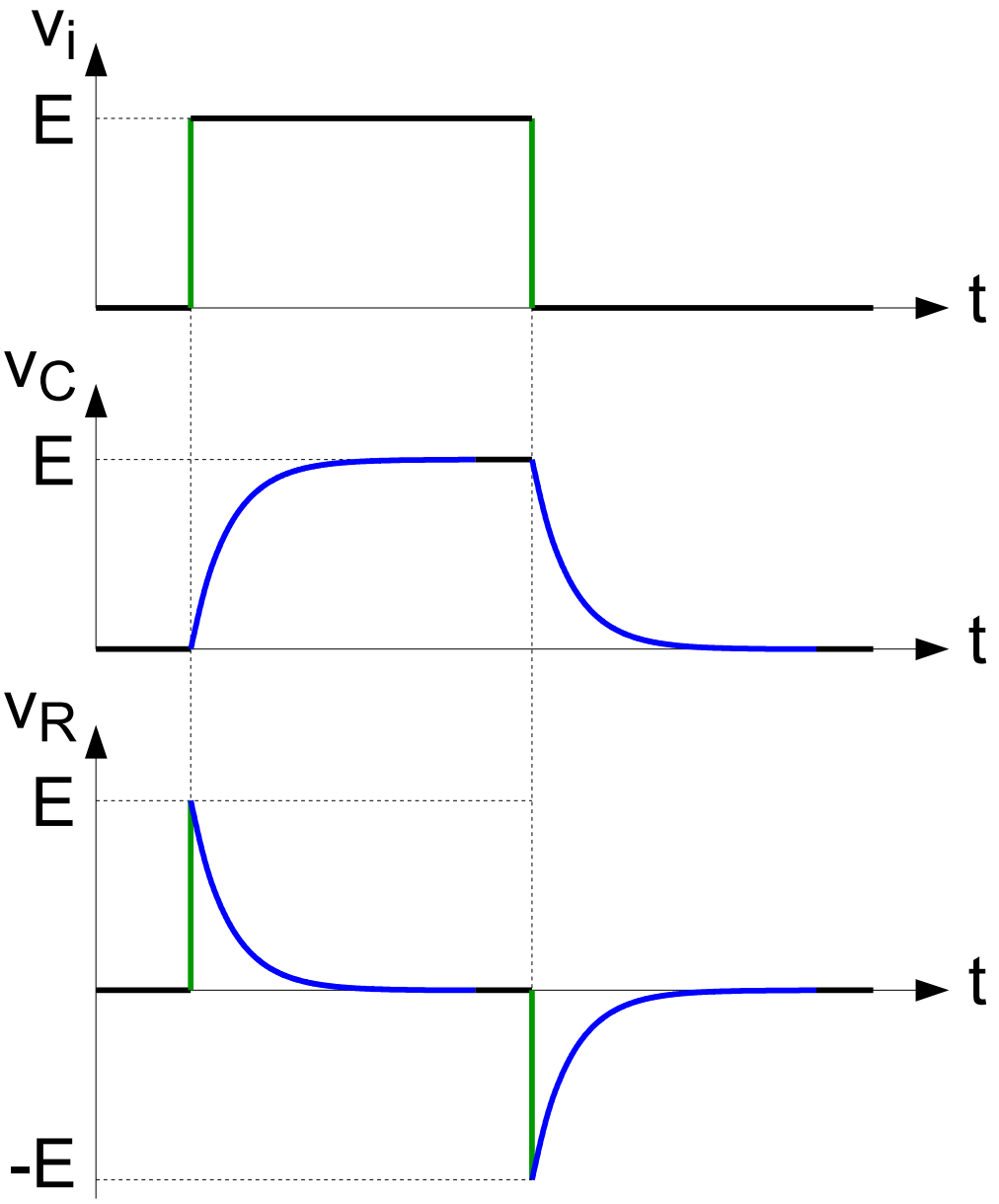
$$\frac{dv_C}{dt} = \frac{v_i - v_C}{\tau}$$

$$v_C(t) = V_{final} + \left(V_{initial} - V_{final}\right) e^{-\frac{t}{\tau}}$$

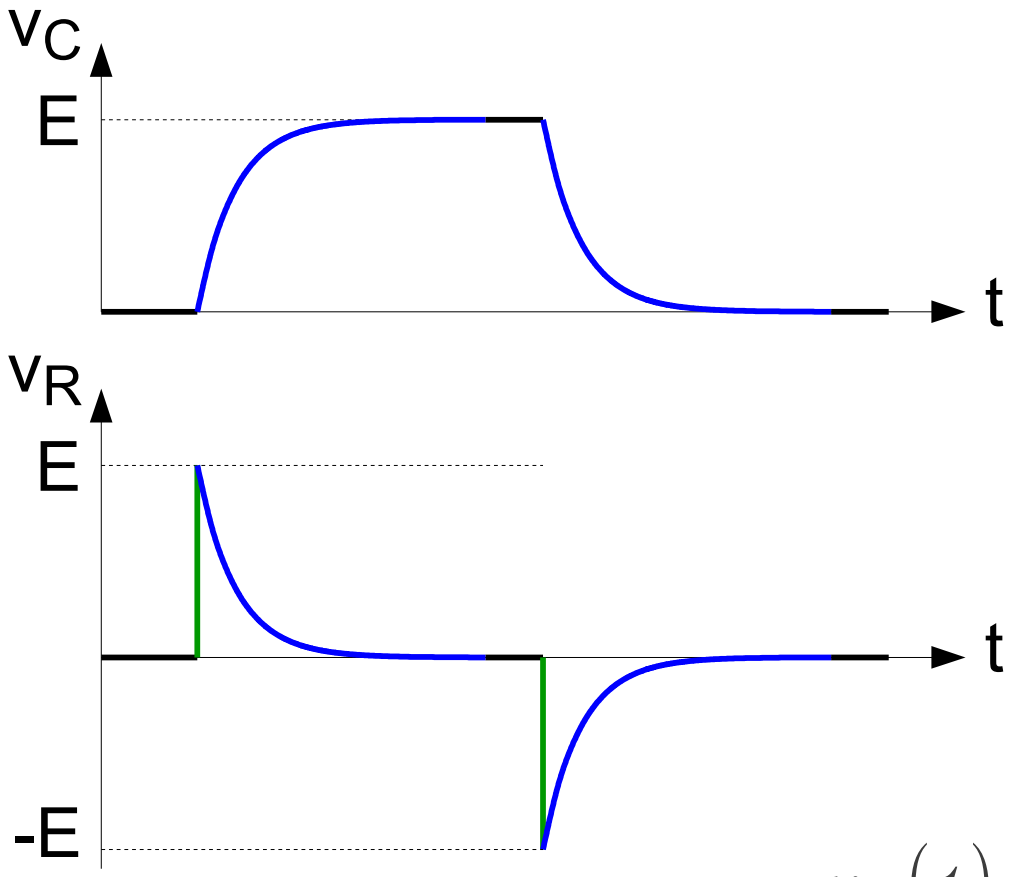
Circuite R-C – analiză (7)







Descriere analitică (1)

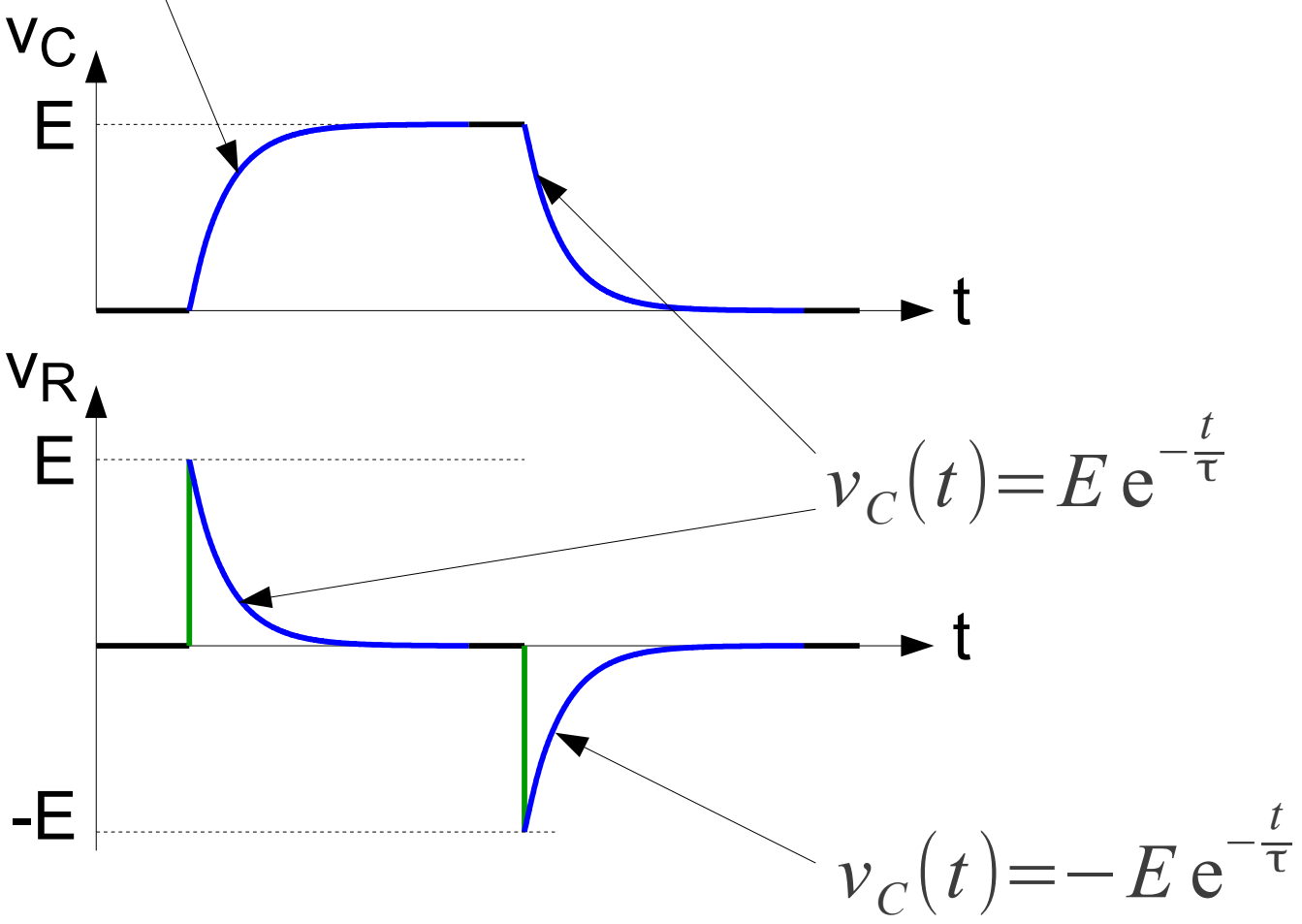


Toate regimurile tranzitorii sunt descrise de ecuația generală (cu deplasarea originii de timp la începutul fenomenului discutat):

$$v_C(t) = V_{final} + (V_{initial} - V_{final})\, e^{-\frac{t}{\tau}}$$

Descriere analitică (2)

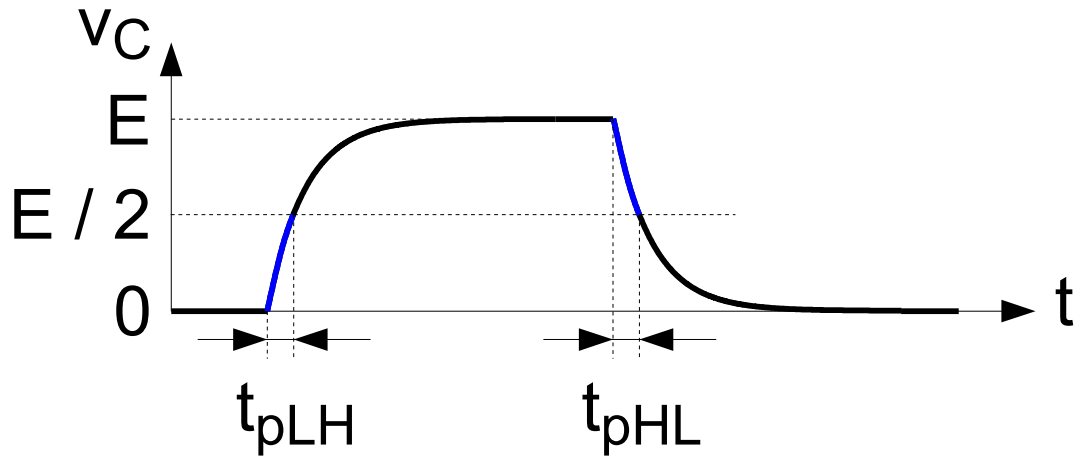
$$v_C(t) = E\left(1 - \mathrm{e}^{-\frac{t}{\tau}}\right)$$



$$v_C(t) = E\,\mathrm{e}^{-\frac{t}{\tau}}$$

$$v_C(t) = -E\,\mathrm{e}^{-\frac{t}{\tau}}$$

Calculul timpilor de propagare



considerăm un comportament simetric => prag logic la E/2

$$v_C(t) = E\left(1 - \mathrm{e}^{-\frac{t}{\tau}}\right)$$

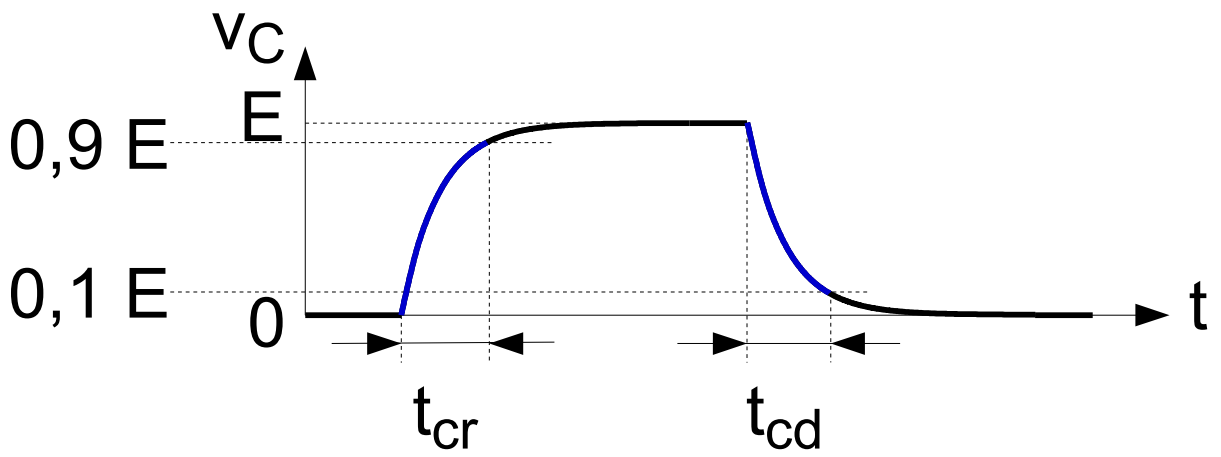
$$\frac{E}{2} = E\left(1 - \mathrm{e}^{-\frac{t_{pLH}}{\tau}}\right)$$

$$v_C(t) = E\,\mathrm{e}^{-\frac{t}{\tau}}$$

$$\frac{E}{2} = E\,\mathrm{e}^{-\frac{t_{pHL}}{\tau}}$$

$$t_{pLH} = t_{pHL} = \tau \ln 2$$

Calculul timpilor de comutație



parcurgerea a 90% din amplitudinea comutației

$$v_C(t) = E\left(1 - \mathrm{e}^{-\frac{t}{\tau}}\right)$$

$$v_C(t) = E\,\mathrm{e}^{-\frac{t}{\tau}}$$

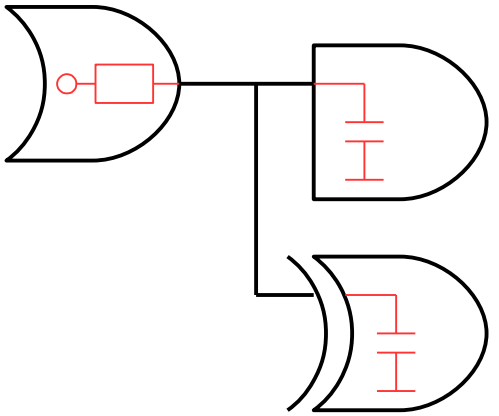
$$\frac{9}{10}E = E\left(1 - \mathrm{e}^{-\frac{t_{cr}}{\tau}}\right)$$

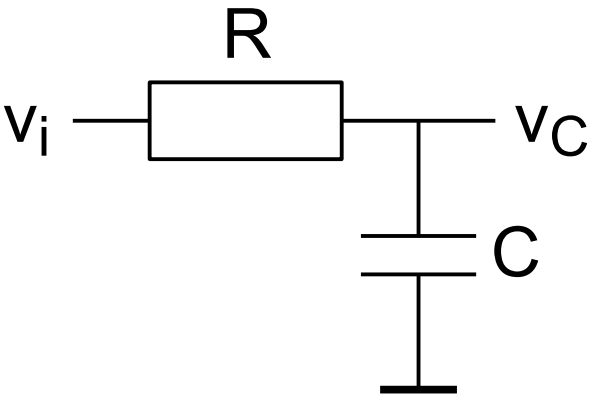
$$\frac{E}{10} = E\,\mathrm{e}^{-\frac{t_{cd}}{\tau}}$$

$$t_{cr} = t_{cd} = \tau \ln 10$$

Consecințe (1)

- Niciun nod nu poate comuta instantaneu
  - datorită capacității parazite
- Fiecare poartă introduce un timp de propagare
- Pentru o comutație rapidă trebuie:
  - rezistență mică
    - consum mare
  - capacitate mică
    - limitare *fan-out*
    - miniaturizare



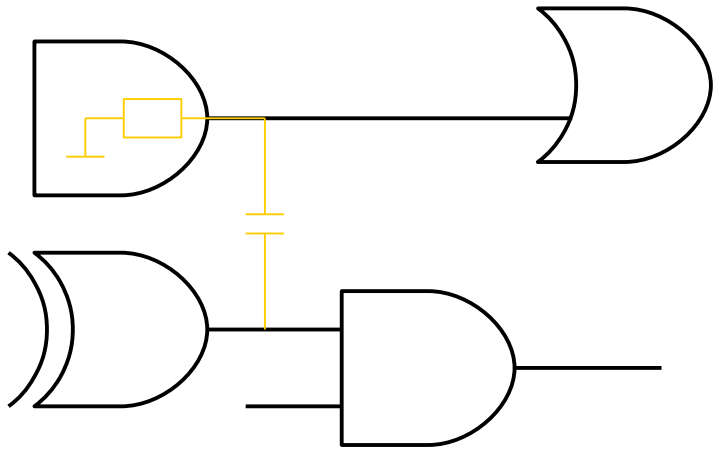


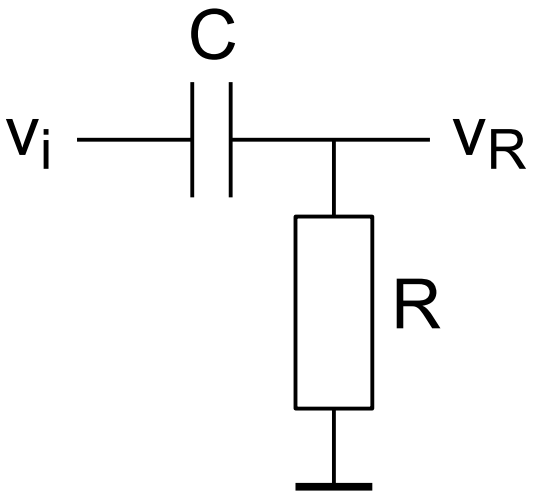
Consecințe (2)

- Circuitele digitale sunt zgomotoase

  (produc perturbații)

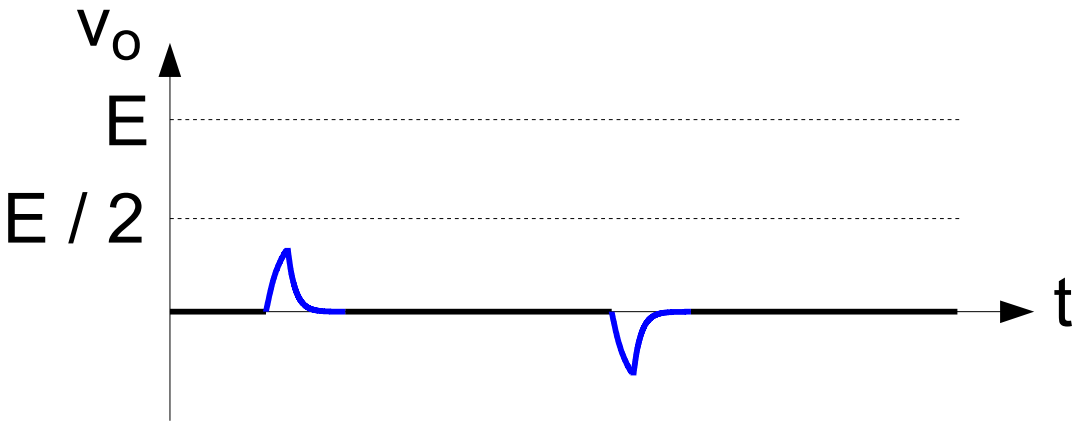
  - dar sunt și tolerante la zgomot

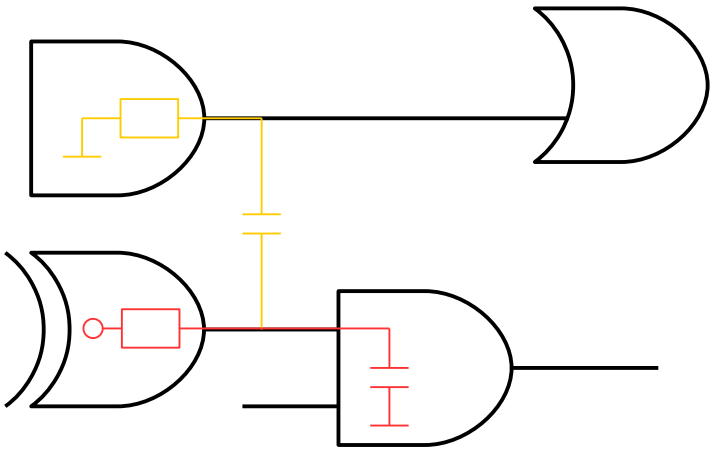


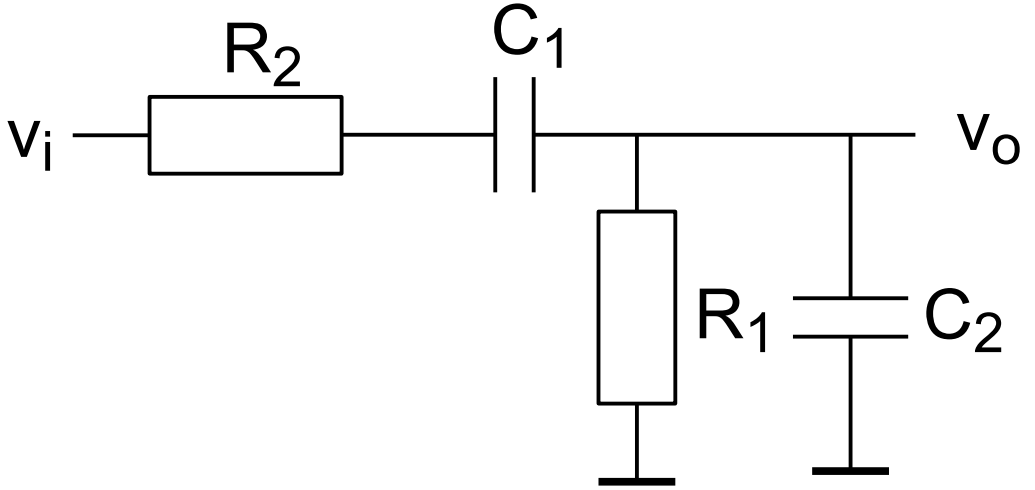


Consecințe (3)

- Model mai elaborat







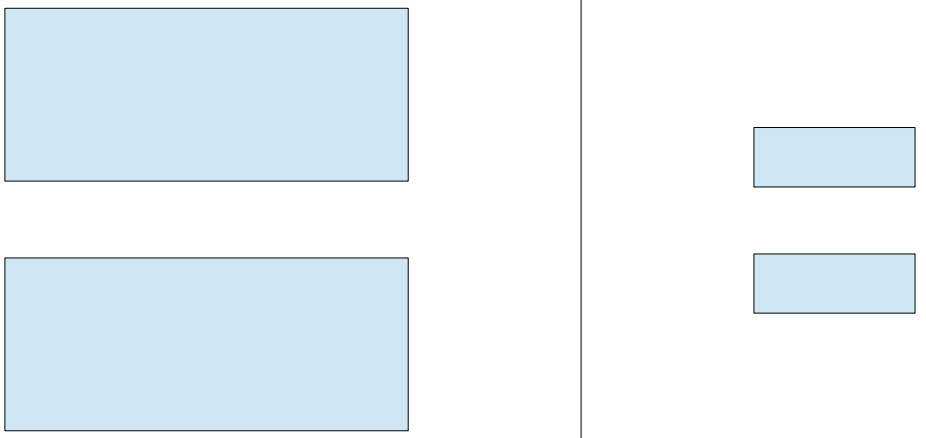
Miniaturizarea circuitelor integrate

- circuitele digitale **nu** sunt aplicații de precizie
  - trebuie doar să distingă între stări
  - pot fi miniaturizate (reduse iterativ dimensiunile)
    - reducerea capacităților parazite
    - sporirea funcționalităților integrate

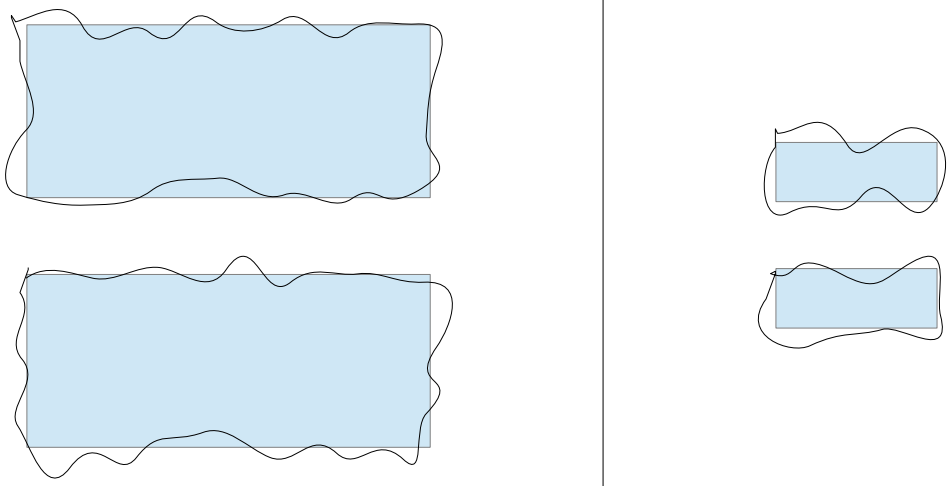
- circuitele analogice **au** cerințe de precizie
  - este mai dificilă miniaturizarea

Exemplificare: rezistențe “egale”

ideal
(desen)



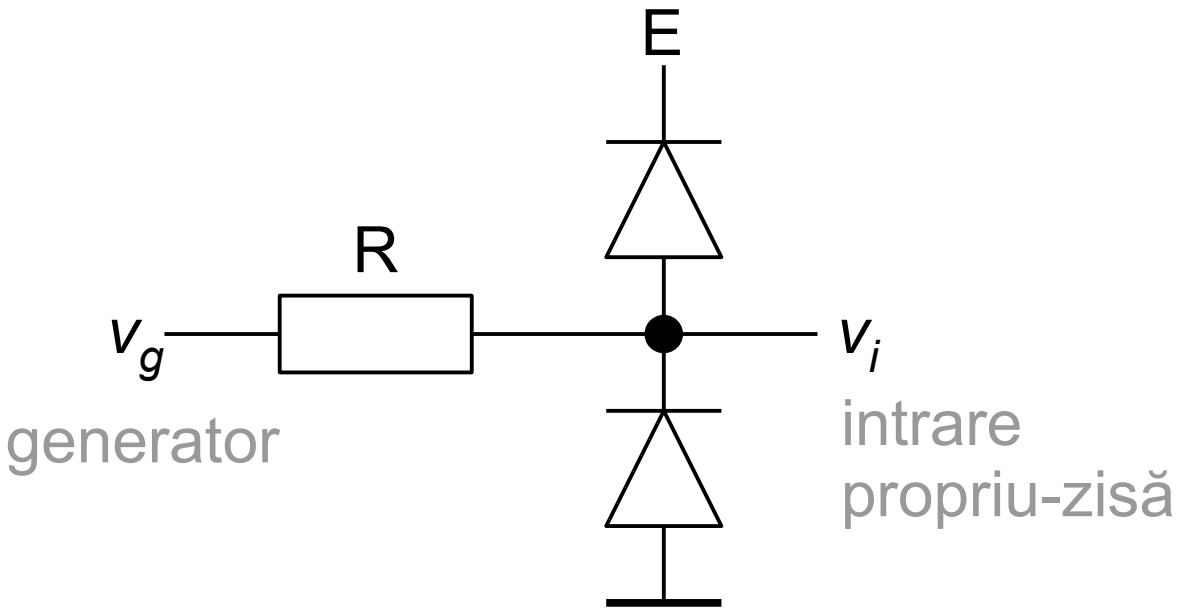
real
(cu imprecizie
de fabricație)

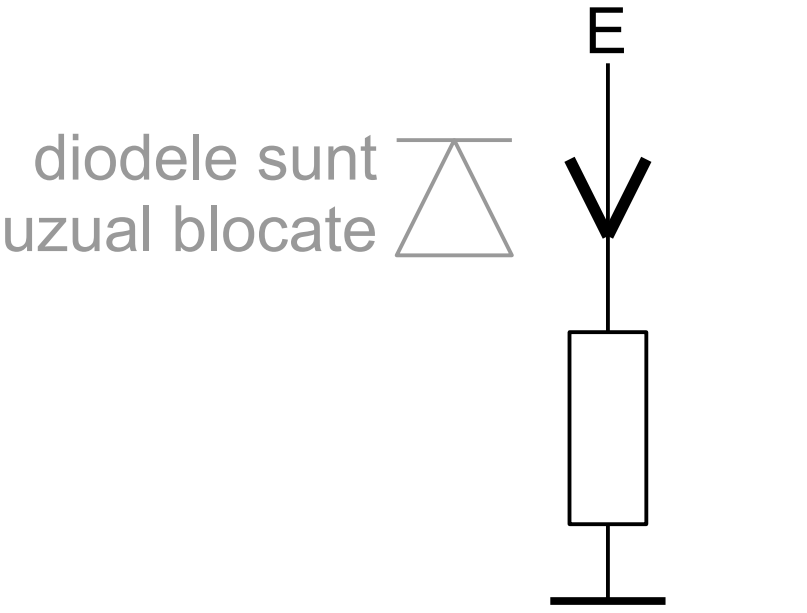


Circuite cu diode

- Limitatoare
  - protecție la supratensiune
  - descărcări electrostatice (*ESD*)

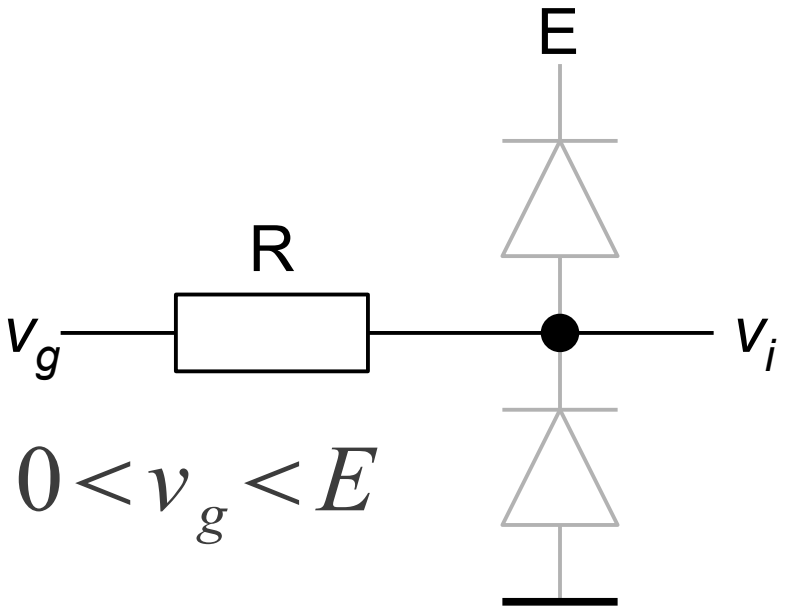
Limitatoare (1)



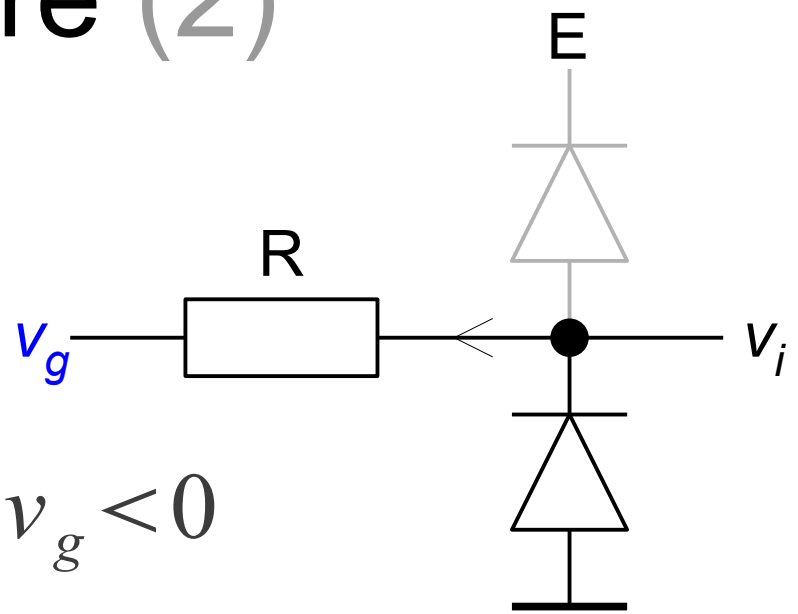


OBS: sensul “natural”
al curentului

Limitatoare (2)

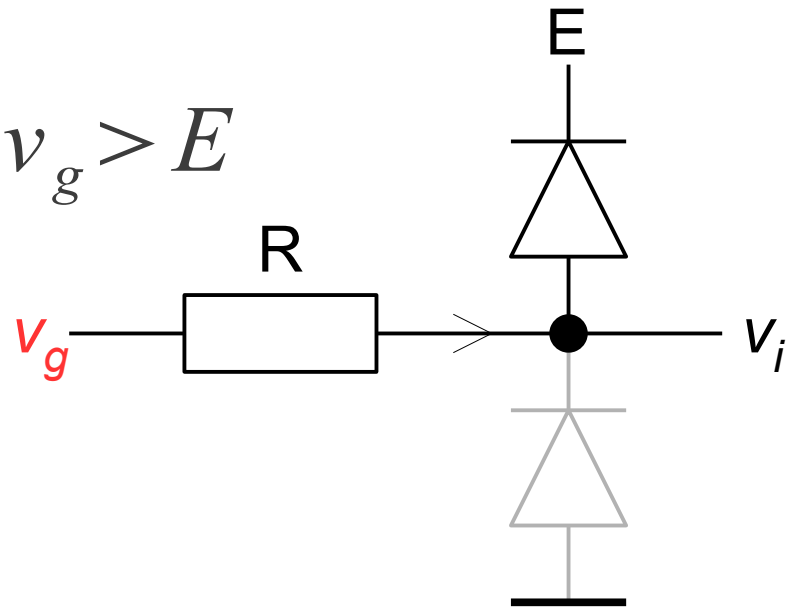


$0 < v_g < E$



$v_g < 0$

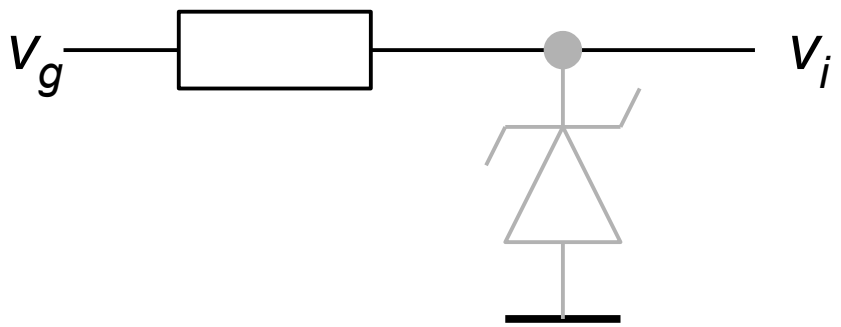
$v_g > E$



limitatorul asigură

$-V_D < v_i < E + V_D$

Limitatoare (3)



diodă Zener uzual blocată;
prin deschidere / străpungere
asigură

$$-V_D < v_i < V_Z$$

Aspecte principale de interes în proiectarea limitatoarelor:

- curentul de avarie, puterea disipată pe rezistență
  - R cât mai mare
- capacitatea parazită a diodelor
  - R cât mai mică